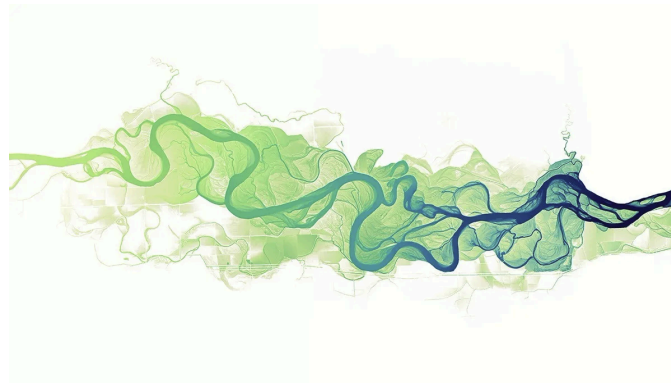


ZÜRICH UNIVERSITY OF APPLIED SCIENCES  
DEPARTEMENT LIFE SCIENCES AND FACILITY MANAGEMENT  
INSTITUTE FOR COMPUTATIONAL LIFE SCIENCES

# **Advanced Earth Observation for Grassland Management:**

## **Improving Automated Mowing Detection Using Multimodal Remote Sensing and Machine Learning**



### **Bachelor's Thesis**

by  
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# **Abstract**

# 1. Introcution

Grassland management at airports is a safety-critical task. Vegetation height and structure on airfield grounds directly influence habitat attractiveness for bird species, which in turn affects the frequency and severity of bird strikes (DeVault et al. 2011). At Zurich Airport, grasslands are subject to regular mowing protocols designed to reduce wildlife hazards. However, monitoring when and where mowing occurs has traditionally relied on manual records created by field staff, a process that is labour-intensive and prone to spatial inaccuracies. To investigate the relationship between mowing practices and bird population dynamics, consistent and spatially reliable mowing records are required.

Satellite-based remote sensing offers a scalable alternative to manual monitoring. The Sentinel-2 constellation, operated by the European Space Agency, provides multispectral imagery at 10 to 20 metre spatial resolution with a revisit cycle of approximately five days (Drusch et al. 2012). Vegetation indices derived from these data, such as the Normalized Difference Vegetation Index (NDVI) and the Normalized Difference Infrared Index (NDII), capture the spectral response of vegetation to management events like mowing, where the abrupt removal of biomass produces a measurable drop in reflectance values (Pettorelli et al. 2005; Andreatta et al. 2022).

In a preceding project work, a pixel-based machine learning workflow for automated mowing detection at Zurich Airport using Sentinel-2 imagery was developed (Klaver 2025). That study demonstrated that standard classifiers, particularly the Support Vector Machine, can detect mowing events with high accuracy on controlled test samples (F1-Score of 0.923). However, the work also identified two fundamental limitations that prevent the system from functioning as a reliable operational tool.

The first limitation concerns data availability. Optical sensors depend on clear atmospheric conditions, and in the temperate climate of central Switzerland, frequent cloud cover regularly interrupts the temporal chain of observations needed for change detection. This problem is well documented in the literature. De Vroey et al. (2021) highlight that relying solely on optical data is particularly problematic in regions where cloud cover is persistent during the growing season. Synthetic Aperture Radar (SAR), which operates independently of weather and illumination conditions, has been proposed as a complementary data source. Several studies have demonstrated that interferometric coherence derived from SAR acquisitions can detect vegetation changes such as mowing, because the removal of biomass alters the scattering properties of the surface (Tamm et al. 2016; Zalite et al. 2016; Reinermann et al. 2022).

The second limitation concerns spatial consistency. The pixel-based classification approach produces predictions independently for each pixel, resulting in spatially fragmented output with characteristic “salt-and-pepper” noise. In the full-scene application conducted in the previous study, models that performed well on test metrics produced scattered false positives across the scene and failed to reconstruct the contiguous shapes of mowing events (Klaver 2025). Spatial postprocessing techniques, such as morphological filtering and superpixel-based segmentation, can address this issue by incorporating neighbourhood context into the classification output. These methods have been applied in remote sensing to improve the spatial coherence of pixel-based classifications, though their specific application to mowing detection remains under-explored. This Bachelor's thesis builds directly on the workflow established in the previous project work and aims to address both of these limitations. **[REWORK THIS PARAGRAPH WITH SOURCES!!]**

The main objectives are:

1. To investigate the integration of Synthetic Aperture Radar (SAR) data from the Sentinel-1 mission as a complementary input to the existing optical feature set, with the goal of reducing dependence on cloud-free optical acquisitions.
2. To implement and evaluate spatial postprocessing methods that improve the spatial consistency of pixel-based predictions.
3. To benchmark the enhanced workflow against the existing baseline model from the previous study, assessing whether the additions yield measurable improvements in detection robustness and operational reliability.

## 2. Literature Review

This chapter reviews the theoretical foundations relevant to the two main methodological extensions introduced in this thesis: the integration of Synthetic Aperture Radar (SAR) data and the application of spatial postprocessing techniques. For a detailed treatment of optical remote sensing with Sentinel-2, vegetation index theory, and the pixel-based machine learning workflow that serves as the baseline for this work, please refer to the preceding project work (Klaver 2025).

### 2.1. Optical Remote Sensing for Grassland Monitoring

Satellite-based optical remote sensing enables continuous observation of vegetation dynamics across large spatial and temporal scales. The Sentinel-2 mission provides multispectral imagery in 13 spectral bands spanning visible, near-infrared, and short-wave infrared wavelengths, with spatial resolutions between 10 and 60 metres and a revisit cycle of approximately five days (Drusch et al. 2012). Vegetation indices derived from these bands quantify biophysical properties of vegetation. The Normalized Difference Vegetation Index (NDVI), which exploits the contrast between red absorption and near-infrared reflectance by chlorophyll-bearing tissue, is the most widely used indicator of vegetation health and biomass (Pettorelli et al. 2005). The Normalized Difference Infrared Index (NDII), which incorporates the short-wave infrared band sensitive to leaf water content, has been shown to yield high accuracy for mowing frequency detection, in some cases outperforming NDVI-based approaches (Andreatta et al. 2022).

Multiple studies have investigated automated mowing detection from optical time series using both rule-based and machine learning approaches. Reinermann et al. (2022) applied a threshold-based algorithm to Sentinel-2 Enhanced Vegetation Index (EVI) time series in Germany. Machine learning classifiers such as Random Forest and Support Vector Machines have also been employed for this task, with Garioud et al. (2019) noting that simpler models can be highly effective for event detection when training data is limited. A comprehensive review of remote sensing approaches for grassland production and management, including mowing detection, is provided by Reinermann et al. (2020).

Despite the effectiveness of optical approaches, their operational reliability is constrained by atmospheric conditions. In temperate climates, cloud cover frequently disrupts the temporal continuity of observations required for change detection (De Vroey et al. 2021). This limitation motivates the integration of weather-independent data sources such as SAR.

### 2.2. Synthetic Aperture Radar

**[Note: This subsection needs a foundational SAR reference, ideally a textbook or ESA technical document, to cite for the basic principles explained below. Search for: “SAR remote sensing fundamentals textbook” or the ESA Sentinel-1 User Guide.]**

Unlike optical sensors, which measure reflected sunlight in the visible and infrared spectrum, SAR is an active imaging system that transmits microwave pulses and records the backscattered signal. Because SAR operates at microwave wavelengths, it is independent of solar illumination and can penetrate cloud cover, making it a valuable complement to optical sensors in regions with frequent atmospheric obstruction.

A SAR sensor emits coherent electromagnetic radiation and measures both the intensity (amplitude) and the phase of the returned signal. The amplitude of the backscattered signal depends on the geometric and dielectric properties of the surface, including roughness, moisture content, and

vegetation structure. SAR data products differ in the level of information they preserve. Ground Range Detected (GRD) products contain only amplitude information, while Single Look Complex (SLC) products retain both amplitude and phase. The phase information in SLC data is essential for interferometric analysis.

**[Note: The above paragraph describes general SAR principles from domain knowledge. A citable source (textbook or technical reference) is needed here. Consider also including a conceptual figure showing the difference between optical and SAR imaging, either self-made or from a cited source.]**

**## Interferometric Coherence** When two SAR acquisitions are taken from nearly the same orbital position at different times (a repeat-pass configuration), their phase signals can be compared through a technique known as SAR interferometry (InSAR). The interferometric coherence is a measure of the similarity between these two complex signals, expressed as a value between 0 and 1. A coherence value of 1 indicates that the scattering properties of the surface remained identical between the two acquisitions, while a value of 0 indicates complete decorrelation.

**[Note: A mathematical definition of coherence (the standard formula with complex cross-correlation) would strengthen this section. Needs a citable source for the formula. Tamm et al. (2016) or Zalite et al. (2016) may contain it, but verify from the full text.]**

For vegetated surfaces, temporal decorrelation is driven primarily by changes in the physical structure of the canopy. Growing vegetation continuously alters its geometry through leaf movement, growth, and water content changes, which causes progressive phase decorrelation and results in lower coherence values. Conversely, when vegetation is removed through mowing, the exposed soil or short stubble presents a more temporally stable scattering surface, leading to higher coherence values in subsequent interferometric pairs.

Tamm et al. (2016) investigated this relationship using Sentinel-1 interferometric coherence calculated from 12-day image pairs over agricultural grasslands. Their results showed that both VH and VV polarisation coherence values were statistically significantly higher after mowing events compared to pre-mowing conditions. The study also found that shorter intervals between the mowing event and the first interferometric acquisition produced higher coherence values, and that coherence tended to remain elevated even 24 to 36 days after mowing. However, precipitation was found to decrease coherence, which can impede mowing detection. Additionally, the 12-day temporal baseline proved insufficient during rapid growth phases, as regrowth between acquisitions reduced the coherence contrast.

Zalite et al. (2016) examined agricultural grassland monitoring using X-band repeat-pass interferometric SAR and similarly found that interferometric coherence was sensitive to management-related changes in vegetation structure. **[Note: Zalite et al. (2016) uses X-band, not C-band like Sentinel-1. If you have access to the full paper, check whether they discuss band-specific differences. This distinction could be worth a sentence. If not, the current formulation is safe based on the abstract.]**

### 2.3. SAR for Mowing Detection

De Vroey et al. (2021) conducted a quantitative assessment of Sentinel-1 C-band SAR for mowing detection using a large in-situ reference dataset. Their results showed that 54% of mowing events could be detected in hay meadows based on coherence jumps. Parcels with a single mowing event during the summer were identified with the highest accuracy at 71%. The study also identified grazing events as a major confounding factor, as most false detections occurred in pastures. The

authors concluded that combining Sentinel-1 and Sentinel-2 data could further improve detection performance.

Reinermann et al. (2022) tested an integrated approach combining Sentinel-2 and Sentinel-1 parameters for mowing detection across Germany. Their optical-only approach using the Enhanced Vegetation Index achieved an F1-Score of 0.64, detecting 60.3% of mowing events. When SAR-based features were investigated during optical data gaps, Polarimetric SAR (PolSAR) entropy proved most reliable for event detection. However, the combined S2+S1 approach, while detecting additional events, also increased false positive detections, resulting in a lower F1-Score of 0.61 for the combined model compared to 0.65 for the optical-only model in the focus region. Komisarenko et al. (2022) developed a deep learning model with an integrated reject option for grassland mowing detection using both Sentinel-1 and Sentinel-2 time series. Applied to approximately 2000 fields in Estonia, the model achieved an event accuracy of 73.3% and an end-of-season accuracy of 94.8%.

**[Note: You could use 1-2 more sources here that specifically discuss coherence-based mowing detection with Sentinel-1 SLC data. Forward-cite Tamm et al. (2016) in Google Scholar to find recent work building on their findings.]**

## 2.4. Spatial Postprocessing Methods

**[Note: This entire section needs literature sources. The concepts described below are accurate but currently unsupported by citations. Priority searches: (1) a foundational OBIA reference such as Blaschke (2010), (2) a paper on SLIC superpixels applied to remote sensing, (3) a paper on morphological postprocessing for classification maps. See keyword suggestions above.]**

Pixel-based classification methods treat each pixel independently, without considering the spatial context of neighbouring pixels. While this approach is computationally efficient, it frequently produces spatially fragmented classification maps characterised by isolated misclassified pixels. In remote sensing, this phenomenon is commonly referred to as “salt-and-pepper” noise. Several families of postprocessing techniques have been developed to address this limitation.

## 2.5. Morphological Operations

Mathematical morphology provides a set of spatial operations that modify the shape and structure of regions in binary or categorical images. The two fundamental operations are erosion, which removes isolated pixels and shrinks the boundaries of classified regions, and dilation, which fills small gaps and expands region boundaries. Combinations of these operations, known as opening (erosion followed by dilation) and closing (dilation followed by erosion), can be applied to classification outputs to remove small spurious detections while preserving the overall structure of larger coherent regions. **[Note: A reference to a remote sensing textbook or paper that applies morphological postprocessing to classification outputs would be appropriate here. Search: “morphological filtering post-classification remote sensing”]**

### Superpixel Segmentation

Superpixel algorithms group pixels into perceptually meaningful clusters based on spectral similarity and spatial proximity. The Simple Linear Iterative Clustering (SLIC) algorithm is one of the most widely used methods for this purpose. SLIC generates compact, approximately uniformly sized segments by adapting k-means clustering to operate jointly in the spatial and spectral domain. In the context of classification postprocessing, superpixel segmentation can be

applied to aggregate pixel-level predictions within each segment. A common strategy is majority voting, where all pixels within a superpixel segment are assigned the class label of the majority. This enforces spatial coherence by ensuring that small clusters of misclassified pixels within an otherwise homogeneous region are corrected to match their surroundings. **[Note: Need a source for SLIC (the original paper is Achanta et al., 2012, “SLIC Superpixels Compared to State-of-the-Art Superpixel Methods”) and ideally a paper applying SLIC or superpixel-based postprocessing to satellite image classification.]**

### **Object-Based Image Analysis**

**[Note: Only include this subsection if you actually implement OBIA or plan to. If your postprocessing is limited to morphological operations and SLIC, this section may not be necessary. Including methods you don't use will raise questions from evaluators. Decide based on your final methodology. If included, search for: Blaschke (2010) “Object based image analysis for remote sensing” as the foundational reference.]** Object-Based Image Analysis (OBIA) represents a more comprehensive approach to incorporating spatial context. Rather than classifying individual pixels, OBIA first segments the image into homogeneous objects and then classifies these objects based on their spectral, spatial, and contextual properties. This approach inherently avoids salt-and-pepper noise because the classification unit is the object rather than the pixel. **[Note: 2-3 sentences on how OBIA has been applied in land cover or vegetation classification, with citations.]** ## Summary **[Note: Optional. A brief paragraph (3-4 sentences) summarising the key takeaways from the literature and explicitly connecting them to the methodology of this thesis. This creates a clean transition to the methods chapter. Keep it short.]**

### 3. Methods

## 4. Results

## 5. Discussion

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## Appendices

## A. Raw Data

## **B. Statistical Outputs**